DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture Week 2: Assignment

Expected Utility · Risk Aversion · CRRA & CARA

Instructions

- This assignment covers Lectures 4–6 and the Week 2 Lab. It carries **100 marks** total.
- Show all working for numerical questions – correct answers with no working receive partial credit only.
- Questions marked **[Python]** require a short script or notebook cell; submit your code alongside your written answers.
- Wealth in this assignment is quoted in raw rupees for readability, but you are encouraged to work in Rs. lakh internally for any Python question, exactly as in the Lab, to avoid numerical precision issues at high γ.
- Submit a single PDF (typeset or clearly scanned handwriting) plus any `.py`/`.ipynb` files, via the course submission portal.

Part A – Concepts (20 marks)

A1. **(4 marks)** Explain, in 2–3 sentences, why the St. Petersburg Paradox demonstrates that maximizing expected *monetary value* cannot be the correct description of rational decision-making.

A2. **(6 marks)** For each utility function below, state whether the individual is risk-averse, risk-neutral, or risk-seeking, and justify your answer using the sign of $u''(W)$.

(a) $u(W) = 5W + 3$

(b) $u(W) = \sqrt{W}$

(c) $u(W) = W^2$

(d) $u(W) = -e^{-2W}$

A3. **(4 marks)** State Jensen's Inequality for a concave function. What does the inequality look like instead for a *convex* utility function, and what risk attitude does that correspond to?

A4. **(3 marks)** True or false, with justification: "A risk-neutral individual would never purchase insurance."

A5. **(3 marks)** Utility is only meaningful up to a positive affine transformation: $u(W)$ and $v(W) = c + d \cdot u(W)$ (with $d > 0$) represent *identical* preferences over lotteries. Explain briefly why $A(W)$ and $R(W)$ computed from u and from v must therefore be exactly equal.

Part B – Certainty Equivalent & the Arrow–Pratt Approximation (30 marks)

B1. **(6 marks)** For $u(W) = \sqrt{W}$, a gamble pays Rs. 90 or Rs. 160 with equal probability. Compute the exact certainty equivalent CE and risk premium π by hand.

B2. **(8 marks)** Derive the Arrow–Pratt approximation $\pi \approx \frac{1}{2}A(\bar{W})\sigma^2$ from a second-order Taylor expansion of $u(W)$ around $\bar{W}$, showing every step (this is the same derivation as in Lecture 5 – reproduce it in your own words to demonstrate you understand each step, not just the final formula).

B3. **(4 marks)** Using the Arrow–Pratt approximation, estimate the risk premium for a log-utility investor ($u(W) = \ln W$) with $\bar{W} =$ Rs. 20,00,000 facing a risk with $\sigma =$ Rs. 3,00,000.

B4. **(6 marks)** An investor faces a gamble of $\pm$Rs. 40,000 around a mean wealth of Rs. 10,00,000 (equal probability), and their measured certainty equivalent for this gamble is Rs. 9,99,200 (so $\pi = 800$). Using the Arrow–Pratt approximation, back out this investor's implied absolute risk aversion coefficient $A(\bar{W})$.

B5. **(6 marks)** Suppose the investor in B4 has **power (CRRA)** utility. Using your answer to B4 and the relationship $R(W) = W \cdot A(W)$, find their implied relative risk-aversion parameter γ. What specific utility function does this correspond to?

Part C – CRRA and CARA Properties (25 marks)

C1. **(6 marks)** Consider $u(W) = -1/W$. Compute $A(W)$ and $R(W)$ directly from this form (do not just quote the general CRRA formula). Confirm your answer matches the $\gamma = 2$ case of power utility from Lecture 6, and briefly explain why this must be so given Part A5.

C2. **(6 marks)** For CARA utility, $R(W) = aW$ is increasing in wealth. Explain, in economic terms, what this implies about how a CARA investor's optimal *portfolio share* (fraction of wealth) invested in a risky asset should change as they become wealthier – and contrast this with what CRRA utility would imply instead.

C3. **(8 marks)** Prove the exact CARA result from Lecture 6: for $W = \bar{W} + \varepsilon$ where ε is a rupee-denominated risk independent of $\bar{W}$, show that

$$CE = \bar{W} - \frac{1}{a} \ln \mathbb{E}\left[e^{-a\varepsilon}\right],$$

and hence that the risk premium π does not depend on $\bar{W}$ at all. Show every step.

C4. **(5 marks)** A CRRA investor has $\gamma = 4$. Facing a proportional gamble, their risk premium is $\pi/\bar{W} = 0.8\%$ when $\bar{W} =$ Rs. 8,00,000. Using *only* the exact CRRA scaling property from Lecture 6 (no new utility calculation), find π (in rupees) when $\bar{W} =$ Rs. 40,00,000.

Part D – Computational (15 marks)

D1. **(8 marks) [Python]** Using your Lab code (or a fresh, self-contained script), consider a CARA investor with wealth $\bar{W} =$ Rs. 20 lakh facing a fixed additive gamble of $\pm$Rs. 2 lakh (equal probability). Compute and plot the *exact* (root-found) risk premium π as a function of a, for a ranging from 0.01 to 0.20 (in units of 1/lakh). Comment on the shape of the curve: is it linear in a? Why or why not?

D2. (7 marks) [Python] On the same axes as D1, plot the Arrow–Pratt approximation $\pi \approx \frac{1}{2}a\sigma^2$ for the same range of a. For what range of a is the approximation within 5% of the exact value? Does this match what you would expect, given that the gamble size (Rs. 2 lakh, a 10% swing on Rs. 20 lakh) is held fixed while a increases?

Part E – Integrative Question (10 marks)

E1. (10 marks) In Week 1, you learned that bonds are a fixed, senior claim while equities are a residual, uncapped claim with historically higher average returns. In Week 2, you learned that a risk-averse investor with concave utility values a risky payoff at *less* than its expected value, sacrificing a risk premium to avoid uncertainty. In a paragraph of 150–200 words, explain why a risk-averse investor with CRRA utility would rationally choose to hold *both* bonds and equities, rather than investing entirely in equities to chase the higher expected return. Your answer should reference: (i) diminishing marginal utility of wealth, (ii) the risk premium concept from this week, and (iii) the fixed-vs-residual claim distinction from Week 1.

End of Assignment. Solutions will be released after the submission deadline.